

JEE MAIN REVISION MASTER: MOTION IN A STRAIGHT LINE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 11 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Velocity-Time & Position-Time Graphs	7 PYQs	Graphical Analysis
Moderate	Kinematic Equations & Free Fall	4 PYQs	Kinematics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Velocity-Time & Position-Time Graphs	$v = \frac{dx}{dt}$, $a = \frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt}$, $\int v \, dt = \Delta x$
Kinematic Equations & Free Fall	$v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$, $s_n = u + \frac{a}{2}(2n - 1)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Velocity-Time & Position-Time Graphs (Moderate Yield — 7 PYQs)

Core Principle: Interpretation of motion curves: slope of x-t is velocity, slope of v-t is acceleration, area under v-t is displacement.

■ **Caution / JEE Pitfall:** Area under speed-time curve gives distance (always positive), while area under velocity-time curve gives displacement.

■ **Speed Trick:** When a is given as a function of x , use $a = v(dv/dx)$ rather than (dv/dt) .

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "The velocity of a particle moving along x-axis is given as $V=x^2-5x+4$ (in m/s) where x denotes the x-coordinate of the particle ..." [Ans: C]
- **[2023 Shift] Q#4 (MCQ):** "A body is moving along a straight line with initial velocity v_0 . Its acceleration a is constant. After t seconds, its velocity becomes..." [Ans: N/A]
- **[2022 Shift] Q#1 (MCQ):** "The displacement x (in m) of a particle of mass m (in kg) moving in one dimension under the action of a force, is related to time ..." [Ans: N/A]

3.2 Kinematic Equations & Free Fall (Moderate Yield — 4 PYQs)

Core Principle: Uniformly accelerated rectilinear motion, vertical motion under gravity, and displacement in the n th second.

■ **Caution / JEE Pitfall:** Assign a consistent sign convention (e.g. upward positive, downward negative) at the beginning of problem solving.

■ **Speed Trick:** Stopping distance $d \propto u^2$. For an object dropped from rest, distance ratio in successive seconds is 1 : 3 : 5 : 7 (Galileo's odd numbers).

Representative Source PYQs:

- **[2026 Shift] Q#2 (MCQ):** "A car covers the first half of the distance between two places at 40 km/h and another half at 50 km/h. The average speed of the car is..." [Ans: B]
- **[2025 Shift] Q#3 (MCQ):** "Two stones begin to fall from rest from the same height, with the second stone starting to fall t seconds after the first falls..." [Ans: N/A]
- **[2020 Shift] Q#4 (MCQ):** "At a metro station, a girl walks up a stationary escalator in 20s. If she remains stationary on the escalator, then the escalator takes her..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The velocity of a particle moving along x -axis is given as $V = x^2 - 5x + 4$ (in m/s) where x denotes the x -coordinate of the particle in metres. The magnitude of the acceleration of the particle when the velocity of the particle is zero is

- (A) 2 m/s^2
- (B) 3 m/s^2
- (C) Zero **[Correct]**
- (D) 1 m/s^2

Correct Answer: C

Solution Analysis: The velocity is given as a function of position: $v = x^2 - 5x + 4$. We need acceleration when the velocity is zero. First find the positions where $v = 0$: $x^2 - 5x + 4 = 0$. Factorising, $(x-1)(x-4) = 0$. So, $x = 1$ or $x = 4$ m. Now, since velocity is given as a function of x , acceleration is not simply (dv/dx) . We use $a = (dv/dt)$. Using chain rule, $a = (dv/dx) \cdot (dx/dt)$. But $(dx/dt) = v$. So, $a = v(dv/dx)$. Here, when the velocity is zero, $v = 0$. Hence, $a = 0 \cdot (dv/dx) = 0$. So the magnitude of acceleration is 0 m/s^2 . Correct option: C

Benchmark Problem 2: KCET 2026 (Q#2)

A car covers the first half of the distance between two places at 40 km/h and another half at 50 km/h. The average speed of the car is

- (A) 45.00 km/h
- (B) 44.44 km/h **[Correct]**
- (C) 43.14 km/h
- (D) 42.04 km/h

Correct Answer: B

Solution Analysis: For average speed, we use $\text{Average speed} = (\text{Total distance} / \text{Total time})$. Here, the car covers equal distances at two different speeds. Let the total distance be $2d$ km. So, first half distance $= d$ km and second half distance $= d$ km. Time taken for first half: $(d / 40)$. Time taken for second half: $(d / 50)$. Total time: $(d / 40) + (d / 50)$. Now average speed: $\text{Average speed} = (2d / ((d / 40) + (d / 50)))$. Taking d common in the denominator: $\text{Average speed} = (2d / d((1 / 40) + (1 / 50)))$. Cancel d : $\text{Average speed} = 2 / ((1 / 40) + (1 / 50))$.